

AI/ML REQUIREMENTS & USE CASES

College Football Fantasy — Team 4

Project	College Football Fantasy
Team	Cohort C — Team 4
Function	AI/ML
Sprint	Sprint 1 — Foundation
Document Type	AI/ML Requirements & Use Cases
Version	1.0
Status	Draft for Team Review
Prepared By	AI/ML Engineering Function
Date	September 2026

Ready for Team Review

DOCUMENT CONTROL

Item	Details
Document Title	AI/ML Requirements & Use Cases
Product	College Football Fantasy
Team	Cohort C — Team 4
Function	AI/ML
Sprint	Sprint 1 — Foundation
Version	1.0
Status	Draft for Review
Primary Purpose	Define AI/ML opportunities, requirements, data needs, architecture considerations, and MVP scope

Version History

Version	Date	Description	Author
0.1	September 2026	Initial AI/ML requirements draft	AI/ML Function
1.0	September 2026	Complete requirements and use-case definition	AI/ML Function

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1. EXECUTIVE SUMMARY

College Football Fantasy is a season-long fantasy platform that allows users to create fantasy teams using college football players, participate in leagues, earn fantasy points and compete on leaderboards.

The AI/ML function will provide intelligent, data-driven capabilities that can help users make better fantasy decisions while improving engagement and personalization.

The primary AI/ML objective is not to replace the platform's official scoring or ranking logic. Instead, AI/ML should operate as an intelligent decision-support layer that analyzes historical and current football data to generate predictions, recommendations and insights.

- Player performance prediction
- Player recommendation
- Lineup decision support
- Gameweek performance projections
- Player trend analysis
- Personalized fantasy insights
- Anomaly detection
- Future AI-powered user personalization

For Sprint 1, the recommended priority is to establish the AI/ML foundation: define use cases, identify required data, establish features and target variables, define evaluation metrics, document the proposed architecture, and determine the most practical MVP model.

2. PROJECT CONTEXT

The College Football Fantasy platform is designed as a season-long fantasy sports application based on college football players.

The platform is expected to support:

- User registration and authentication
- Fantasy team creation
- Player selection
- Gameweek participation
- Fantasy scoring
- Leagues
- Leaderboards
- Player and team information
- Statistical data
- User engagement and retention

The technical environment identified for the project includes React Native / React, Node.js, PostgreSQL, REST APIs, JWT-based authentication, cloud infrastructure, college football data sources/APIs, validation and scoring

services, and ranking/leaderboard services.

The AI/ML layer should integrate with these existing services without becoming the authoritative source for fantasy scores.

3. PURPOSE OF THE DOCUMENT

This document defines the proposed AI/ML requirements and use cases for the College Football Fantasy platform.

- Identify where AI/ML can provide value.
- Define required data and model inputs.
- Define expected AI/ML outputs.
- Describe application and API integration.
- Define model evaluation and monitoring.
- Address security, privacy and explainability.
- Define the MVP and future roadmap.

4. AI/ML OBJECTIVES

4.1 Improve Fantasy Decision-Making

Provide data-driven insights that help users evaluate player choices and potential lineups.

4.2 Predict Player Performance

Estimate expected fantasy performance using historical and contextual data.

4.3 Recommend Players

Identify players who may represent strong fantasy opportunities.

4.4 Provide Personalized Insights

Eventually provide recommendations based on individual team composition and preferences.

4.5 Detect Unusual Activity

Explore ML-based approaches for identifying unusual patterns that could indicate data problems, scoring anomalies or suspicious platform activity.

4.6 Increase User Engagement

Use intelligent insights to make the fantasy experience more useful and engaging.

4.7 Maintain Trust

Present predictions as predictions or recommendations, not guaranteed outcomes.

5. AI/ML SCOPE

5.1 In Scope

- Data requirements
- Data preparation
- Feature engineering

- Player performance prediction
- Player recommendation
- Lineup decision support
- Gameweek projections
- Model evaluation
- AI/ML API integration
- Model monitoring
- Explainability
- AI/ML security considerations

5.2 Out of Scope for Initial MVP

- Fully autonomous lineup management
- Guaranteed outcome prediction
- Automated fantasy scoring
- Fully autonomous user decisions
- Complex deep-learning systems without sufficient data
- Real-money betting recommendations
- Unverified injury predictions
- Automatic model retraining without validation

6. PROPOSED AI/ML USE CASES

ID	Use Case	Priority	MVP
UC-01	Player Performance Prediction	High	Yes
UC-02	Player Recommendation	High	Yes
UC-03	Lineup Decision Support	High	Yes
UC-04	Gameweek Projection	High	Yes
UC-05	Player Trend Analysis	Medium	Later / MVP extension
UC-06	Personalized Recommendations	Medium	Future
UC-07	Anomaly Detection	Medium	Future
UC-08	Natural-Language Fantasy Assistant	Low	Future
UC-09	Advanced Team Optimization	Medium	Future

7. DETAILED USE-CASE REQUIREMENTS

UC-01 — Player Performance Prediction

The system predicts a player's expected fantasy performance for an upcoming gameweek using historical and contextual information.

Objective: Help users understand the potential fantasy value of players before making roster decisions.

Potential Inputs

- Historical fantasy points
- Recent game performance
- Games played
- Offensive statistics
- Defensive statistics
- Position
- Team performance
- Opponent information
- Recent form
- Home/away context where available
- Historical usage
- Reliable player availability information

Output

- Expected fantasy points
- Prediction range
- Confidence indication
- Supporting factors

UC-02 — Player Recommendation

The system recommends players who may be valuable for a user's fantasy team.

- Projected fantasy points
- Recent performance
- Player consistency
- Position requirements
- Team context
- Opponent
- Player availability
- Current roster
- Remaining roster slots
- User's fantasy strategy

The system should return a ranked list of eligible players with understandable recommendation factors where possible.

UC-03 — Lineup Decision Support

AI/ML can analyze the user's available players and provide suggestions regarding which players could form a strong lineup.

- Eligible players
- Position constraints
- Projected performance
- Player availability

- Recent performance
- Team/opponent context
- Fantasy roster rules

The AI should recommend a lineup rather than silently modify a user's team. The final decision remains with the user.

UC-04 — Gameweek Projection

The system generates expected fantasy performance for players before an upcoming gameweek. Outputs may include expected points, expected performance category, confidence level, recent trend and recommendation status.

UC-05 — Player Trend Analysis

The system identifies changes in player performance over time. Possible categories are Improving, Stable, Declining, and Insufficient Data.

UC-06 — Personalized Recommendations

A future recommendation system could learn from previous player selections, team composition, league participation, roster changes and interaction with recommendations. Sufficient historical user data should exist before deployment.

UC-07 — Anomaly Detection

Future ML capabilities may identify unusual scoring changes, unexpected statistics, repeated unusual account behavior, data ingestion anomalies or unusual fantasy activity. This should complement, not replace, deterministic security and validation controls.

8. FUNCTIONAL REQUIREMENTS

ID	Requirement	Description
FR-01	Data Ingestion	Obtain approved player, team and game statistics from authorized data sources.
FR-02	Data Validation	Validate incoming data before it is used by a model.
FR-03	Data Preprocessing	Clean, transform and prepare raw data for model consumption.
FR-04	Feature Generation	Generate approved model features from historical and current data.
FR-05	Prediction	Generate predictions for supported players and gameweeks.
FR-06	Recommendation	Rank eligible players according to defined criteria.
FR-07	Confidence	Provide an indication of prediction uncertainty where technically appropriate.
FR-08	Explanation	Provide understandable factors contributing to recommendations.
FR-09	Model Versioning	Make each production prediction traceable to a model version.
FR-10	Monitoring	Monitor model performance and prediction quality.
FR-11	Failure Handling	Gracefully fall back to non-AI information when reliable predictions are unavailable.
FR-12	No Authoritative Scoring	AI predictions must not determine official fantasy scores; the deterministic scoring engine remains authoritative.

9. DATA REQUIREMENTS

Data Category	Examples
Player Data	Player ID, position, team
Game Data	Game ID, date, teams, result
Player Statistics	Passing, rushing, receiving and defensive statistics
Fantasy Data	Fantasy points, gameweek points
Team Data	Team performance and offensive/defensive statistics
Opponent Data	Opponent strength and performance
Historical Data	Previous game and season performance
Availability Data	Approved player status information
User Data	Only where required for personalization

Data Quality Requirements

- Missing values
- Duplicate records
- Invalid player identifiers
- Invalid team identifiers
- Incorrect statistical values
- Data type inconsistencies
- Timestamp inconsistencies
- Unexpected distribution changes

AI/ML quality depends heavily on reliable data. A sophisticated model cannot compensate for unreliable or incorrectly processed data.

10. FEATURE ENGINEERING REQUIREMENTS

Player-Level Features

- Average fantasy points
- Recent fantasy points
- Rolling average
- Performance variance
- Games played
- Position
- Recent statistical production

Team-Level Features

- Team offensive performance
- Team defensive performance
- Recent team form
- Team scoring trends

Opponent Features

- Opponent defensive performance
- Opponent offensive performance
- Recent opponent form

Contextual Features

- Gameweek
- Season
- Home/away status
- Recent workload
- Reliable player availability information

Feature selection should be based on data availability, predictive value and the risk of introducing data leakage.

11. MODEL REQUIREMENTS

The initial model should prioritize reliability, interpretability, fast inference, ease of maintenance and adequate predictive performance.

The team should begin with an appropriate baseline model before considering complex architectures.

- Linear regression
- Regularized regression
- Decision trees
- Random forest
- Gradient boosting
- Other suitable supervised-learning models

The final model should be selected based on actual dataset characteristics and evaluation results rather than assuming that greater complexity is automatically better.

Primary Model Target

Expected fantasy points for the upcoming gameweek.

Additional targets may include probability of exceeding a points threshold, player performance category or expected statistical output.

12. MODEL EVALUATION AND PERFORMANCE

Evaluation Metrics

- MAE (Mean Absolute Error): average absolute difference between predicted and actual fantasy points.
- RMSE (Root Mean Squared Error): gives greater weight to larger errors.
- R^2 : measures explained variation.
- Ranking metrics: useful for recommendation quality.

Training and Validation Strategy

Where appropriate, datasets should be split chronologically into training, validation and test datasets. Randomly mixing future information into historical training data should be avoided when it would create unrealistic evaluation

results.

Data Leakage Prevention

When predicting a future gameweek, the model must not use information that would only become available after the prediction target occurs. Chronological validation and careful feature construction are required.

13. AI/ML SYSTEM ARCHITECTURE

Proposed high-level architecture:

College Football Data Sources → Data Ingestion → Data Validation → PostgreSQL / Data Store → Feature Engineering → ML Model → Prediction Service → AI/ML API → Node.js Backend → React / React Native Application → User

The AI/ML system should remain logically separated from the core scoring engine.

14. APPLICATION INTEGRATION REQUIREMENTS

Recommended flow:

User → React Native / React → Node.js Backend → Authentication / Authorization → AI/ML API → Prediction Model → Prediction Response → Node.js Backend → Frontend → User

The backend should act as the primary intermediary between the client application and AI/ML services. The client should not directly access protected model infrastructure.

15. API REQUIREMENTS

Illustrative endpoints:

- GET /api/ai/players/{playerId}/projection
- GET /api/ai/players/recommendations
- GET /api/ai/lineup/recommendation
- GET /api/ai/gameweeks/{gameweekId}/projections

Illustrative Prediction Response

```
{"playerId":"PLAYER_ID","projectedPoints":18.4,"confidence":"medium","modelVersion":"v1.0","generatedAt":"2026-09-11T12:00:00Z"}
```

This is an illustrative structure; final endpoint naming and schemas should follow the backend team's API standards.

16. SECURITY, PRIVACY AND DATA GOVERNANCE

- Authenticate AI/ML API requests.
- Authorize access to AI/ML endpoints.
- Protect model and infrastructure credentials.
- Never expose API keys in frontend code.
- Protect sensitive configuration.
- Validate API inputs.
- Apply rate limiting.

- Log security-relevant events.
- Restrict database access.
- Encrypt sensitive data in transit.
- Apply least-privilege access.
- Monitor abnormal API usage.

Secrets Management

Data-provider API keys, database credentials, cloud credentials and model-service credentials must not be hard-coded into source code, mobile applications, Git repositories or public configuration files. Use an approved secrets-management mechanism.

Privacy

If user behavior is later used for personalization, collect only information necessary for the intended functionality and apply data minimization, access control, retention, transparency and secure storage principles.

17. EXPLAINABILITY AND USER TRUST

AI recommendations should be understandable to users. Rather than simply displaying a recommendation, the platform should eventually show validated contributing factors such as strong recent performance, projected fantasy output or consistent recent production.

Explanations must reflect actual model inputs or validated decision factors and must not invent reasons.

Use language such as Projected points, Estimated performance, Recommended player, Prediction and Confidence. Avoid guaranteed or certain-outcome claims.

18. AI/ML RISKS AND MITIGATIONS

Risk	Impact	Mitigation
Poor data quality	High	Data validation
Insufficient historical data	High	Start with baseline models
Data leakage	High	Chronological validation
Model overfitting	High	Cross-validation / holdout testing
Model drift	Medium/High	Continuous monitoring
Biased recommendations	Medium	Evaluate model behavior
Incorrect predictions	Medium	Display uncertainty
API failure	Medium	Graceful fallback
Excessive complexity	Medium	Start with simple models
Privacy issues	High	Data minimization
Security compromise	High	Authentication, authorization and secrets management
User overreliance	Medium	Clear prediction disclosures

19. MVP RECOMMENDATION

For the first implementation, Team 4 should avoid attempting to build every AI/ML feature simultaneously.

Recommended MVP

- Player Performance Prediction — predict expected fantasy points.
- Player Recommendation — rank players according to projected fantasy value and relevant constraints.
- Basic Lineup Decision Support — use predictions to provide lineup suggestions.
- Gameweek Projection — provide projections for eligible players before each gameweek.

These capabilities provide a coherent AI/ML MVP while remaining technically realistic.

20. SPRINT 1 DELIVERABLES

- AI/ML use-case definition
- Data requirements
- Feature definition
- Prediction target definition
- Model strategy and evaluation methodology
- AI/ML architecture
- Preliminary API contract
- Security requirements and coordination with cybersecurity

Sprint 1 should focus on planning and technical definition rather than prematurely committing to a complex production model.

21. ACCEPTANCE CRITERIA

- ✓ AI/ML objectives have been documented.
- ✓ Major AI/ML use cases have been identified.
- ✓ Each priority use case has a clear purpose.
- ✓ MVP and future features have been separated.
- ✓ Required data sources have been identified.
- ✓ Candidate model features have been documented.
- ✓ Prediction target has been defined.
- ✓ Model evaluation metrics have been identified.
- ✓ Data leakage risks have been addressed.
- ✓ AI/ML architecture has been defined.
- ✓ Backend/API integration requirements have been documented.
- ✓ Security requirements have been documented.
- ✓ Privacy considerations have been addressed.
- ✓ Explainability requirements have been defined.
- ✓ Model monitoring requirements have been defined.
- ✓ AI/ML risks and mitigations have been documented.
- ✓ Sprint 1 AI/ML deliverables are ready for team review.

22. FUTURE AI/ML ENHANCEMENTS

Advanced Player Ranking

More sophisticated ranking models incorporating multiple contextual factors.

Personalized Fantasy Recommendations

Recommendations tailored to individual user behavior and roster composition.

Team Optimization

Optimization algorithms that consider roster constraints and projected performance simultaneously.

Natural-Language Fantasy Assistant

A conversational assistant for questions about player projections and rosters.

Anomaly Detection

Machine learning for detecting unusual data or platform behavior.

Continuous Model Improvement

Controlled retraining as additional historical gameweek data becomes available.

23. CONCLUSION

The AI/ML function for College Football Fantasy — Team 4 will provide an intelligent decision-support layer capable of transforming football and fantasy data into useful predictions and recommendations.

The initial focus should be on player performance prediction, player recommendations, gameweek projections and basic lineup decision support.

Sprint 1 should establish the foundation required to implement these capabilities successfully, including data requirements, feature definitions, model strategy, evaluation metrics, architecture, API integration, security, privacy and monitoring.

The AI/ML system should be accurate, explainable, secure, scalable and maintainable, while clearly communicating that predictions are estimates rather than guaranteed outcomes.

This approach gives Team 4 a realistic AI/ML roadmap that can evolve as more College Football Fantasy data becomes available throughout the season.

APPROVAL

Role	Name	Status	Date
AI/ML Function	_____	Pending	_____
Product Manager	Aliyu Muhammed	Pending	_____
Technical Lead	_____	Pending	_____
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